



HK IMO Training 2019-2020
Problem Set 4
For 26 October 2019



Problem 13. Two circles ω_1 and ω_2 with centers O_1 and O_2 respectively intersect each other at points A and B , and point O_1 lies on ω_2 . Let P be an arbitrary point lying on ω_1 . Lines BP , AP and O_1O_2 cut ω_2 for the second time at points X , Y and C , respectively. Prove that quadrilateral $XPYC$ is a parallelogram.

Problem 14. Let $ABCD$ be a parallelogram and let K be a point on line AD such that $BK = AB$. Suppose that P is an arbitrary point on AB , and the perpendicular bisector of PC intersects the circumcircle of $\triangle APD$ at points X, Y . Prove that the circumcircle of $\triangle ABK$ passes through the orthocentre of $\triangle AXY$.

Problem 15. Consider an 8×8 chessboard with all 64 unit squares coloured white. We colour 12 unit squares black in an arbitrary manner. Prove that we can always find four rows and four columns containing the 12 black unit squares.

Problem 16. Determine whether there exists a finite set S of prime numbers such that, for every integer $n \geq 2$, the number $2^2 + 3^2 + \dots + n^2$ is divisible by some prime number $p \in S$.